

Table of contents

\\Research

<table border="1"><tr><td>Investigators</td></tr></table>	Investigators
Investigators	

<table border="1"><tr><td><table border="1"><tr><td>XChen</td></tr></table></td></tr></table>	<table border="1"><tr><td>XChen</td></tr></table>	XChen
<table border="1"><tr><td>XChen</td></tr></table>	XChen	
XChen		

<table border="1"><tr><td><table border="1"><tr><td>31P CSI 20230503 NOE</td></tr></table></td></tr></table>	<table border="1"><tr><td>31P CSI 20230503 NOE</td></tr></table>	31P CSI 20230503 NOE
<table border="1"><tr><td>31P CSI 20230503 NOE</td></tr></table>	31P CSI 20230503 NOE	
31P CSI 20230503 NOE		

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>localizer</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>localizer</td></tr></table></td></tr></table>	<table border="1"><tr><td>localizer</td></tr></table>	localizer
<table border="1"><tr><td><table border="1"><tr><td>localizer</td></tr></table></td></tr></table>	<table border="1"><tr><td>localizer</td></tr></table>	localizer	
<table border="1"><tr><td>localizer</td></tr></table>	localizer		
localizer			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>AAHead_Scout</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>AAHead_Scout</td></tr></table></td></tr></table>	<table border="1"><tr><td>AAHead_Scout</td></tr></table>	AAHead_Scout
<table border="1"><tr><td><table border="1"><tr><td>AAHead_Scout</td></tr></table></td></tr></table>	<table border="1"><tr><td>AAHead_Scout</td></tr></table>	AAHead_Scout	
<table border="1"><tr><td>AAHead_Scout</td></tr></table>	AAHead_Scout		
AAHead_Scout			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>T1_MEMPRAGE_64ch</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>T1_MEMPRAGE_64ch</td></tr></table></td></tr></table>	<table border="1"><tr><td>T1_MEMPRAGE_64ch</td></tr></table>	T1_MEMPRAGE_64ch
<table border="1"><tr><td><table border="1"><tr><td>T1_MEMPRAGE_64ch</td></tr></table></td></tr></table>	<table border="1"><tr><td>T1_MEMPRAGE_64ch</td></tr></table>	T1_MEMPRAGE_64ch	
<table border="1"><tr><td>T1_MEMPRAGE_64ch</td></tr></table>	T1_MEMPRAGE_64ch		
T1_MEMPRAGE_64ch			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table>	<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap
<table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table>	<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap	
<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap		
fastestmap			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table>	<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap
<table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table>	<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap	
<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap		
fastestmap			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table>	<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap
<table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table>	<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap	
<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap		
fastestmap			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table>	<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap
<table border="1"><tr><td><table border="1"><tr><td>fastestmap</td></tr></table></td></tr></table>	<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap	
<table border="1"><tr><td>fastestmap</td></tr></table>	fastestmap		
fastestmap			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>SPECIAL_ACC</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>SPECIAL_ACC</td></tr></table></td></tr></table>	<table border="1"><tr><td>SPECIAL_ACC</td></tr></table>	SPECIAL_ACC
<table border="1"><tr><td><table border="1"><tr><td>SPECIAL_ACC</td></tr></table></td></tr></table>	<table border="1"><tr><td>SPECIAL_ACC</td></tr></table>	SPECIAL_ACC	
<table border="1"><tr><td>SPECIAL_ACC</td></tr></table>	SPECIAL_ACC		
SPECIAL_ACC			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>head_csi_fid</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>head_csi_fid</td></tr></table></td></tr></table>	<table border="1"><tr><td>head_csi_fid</td></tr></table>	head_csi_fid
<table border="1"><tr><td><table border="1"><tr><td>head_csi_fid</td></tr></table></td></tr></table>	<table border="1"><tr><td>head_csi_fid</td></tr></table>	head_csi_fid	
<table border="1"><tr><td>head_csi_fid</td></tr></table>	head_csi_fid		
head_csi_fid			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>head_csi_fid</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>head_csi_fid</td></tr></table></td></tr></table>	<table border="1"><tr><td>head_csi_fid</td></tr></table>	head_csi_fid
<table border="1"><tr><td><table border="1"><tr><td>head_csi_fid</td></tr></table></td></tr></table>	<table border="1"><tr><td>head_csi_fid</td></tr></table>	head_csi_fid	
<table border="1"><tr><td>head_csi_fid</td></tr></table>	head_csi_fid		
head_csi_fid			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>Ax</td><td>TSE</td><td>PD</td><td>T2</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>Ax</td><td>TSE</td><td>PD</td><td>T2</td></tr></table></td></tr></table>	<table border="1"><tr><td>Ax</td><td>TSE</td><td>PD</td><td>T2</td></tr></table>	Ax	TSE	PD	T2
<table border="1"><tr><td><table border="1"><tr><td>Ax</td><td>TSE</td><td>PD</td><td>T2</td></tr></table></td></tr></table>	<table border="1"><tr><td>Ax</td><td>TSE</td><td>PD</td><td>T2</td></tr></table>	Ax	TSE	PD	T2	
<table border="1"><tr><td>Ax</td><td>TSE</td><td>PD</td><td>T2</td></tr></table>	Ax	TSE	PD	T2		
Ax	TSE	PD	T2			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>t2_tse_dark-fluid_tra</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>t2_tse_dark-fluid_tra</td></tr></table></td></tr></table>	<table border="1"><tr><td>t2_tse_dark-fluid_tra</td></tr></table>	t2_tse_dark-fluid_tra
<table border="1"><tr><td><table border="1"><tr><td>t2_tse_dark-fluid_tra</td></tr></table></td></tr></table>	<table border="1"><tr><td>t2_tse_dark-fluid_tra</td></tr></table>	t2_tse_dark-fluid_tra	
<table border="1"><tr><td>t2_tse_dark-fluid_tra</td></tr></table>	t2_tse_dark-fluid_tra		
t2_tse_dark-fluid_tra			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>resolve_4scan_trace_tra_p2_192</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>resolve_4scan_trace_tra_p2_192</td></tr></table></td></tr></table>	<table border="1"><tr><td>resolve_4scan_trace_tra_p2_192</td></tr></table>	resolve_4scan_trace_tra_p2_192
<table border="1"><tr><td><table border="1"><tr><td>resolve_4scan_trace_tra_p2_192</td></tr></table></td></tr></table>	<table border="1"><tr><td>resolve_4scan_trace_tra_p2_192</td></tr></table>	resolve_4scan_trace_tra_p2_192	
<table border="1"><tr><td>resolve_4scan_trace_tra_p2_192</td></tr></table>	resolve_4scan_trace_tra_p2_192		
resolve_4scan_trace_tra_p2_192			

\\Research\Investigators\XChen\31P CSI 20230503 NOE\localizer

TA: 19 sec Coil Selection: Auto Voxel Size: 0.5×0.5×10.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
TE	5.00 ms
Averages	1
Concatenations	7
AutoAlign	---

Contrast - Common

TR	20.0 ms
TE	5.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	40 deg

Contrast - Common

Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
Base Resolution	256
Phase Resolution	50 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Image Based
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal

Geometry - Common

Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	7

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
---------------------	----------

System - Adjustments

B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.0 ms
Segments	1
Concatenations	7

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	280 mm
FOV Phase	100.0 %
Phase Resolution	50 %

Physio - PACE

Resp. Control	Off
Concatenations	7

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	180 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators\XChen\31P CSI 20230503 NOE\AAHead_Scout

TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
TE	1.37 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.15 ms
TE	1.37 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.2 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Noise Masking	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.15 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\Research\Investigators\XChen\31P CSI 20230503 NOET1_MEMPRAGE_64ch

TA: 6:30 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	60 %
Slice Resolution	100 %
Interpolation	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain
Coil Elements	HC1-7

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2530.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Contrast - Common

TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Magn. Preparation	Non-sel. IR
TI	1100 ms
Flip Angle	7 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	4
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R2.2 P1.7 F31.8
R	2.2 mm
P	1.7 mm
F	31.8 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	32 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00

System - Tx/Rx

Image Scaling	1.000
---------------	-------

Physio - Signal

1st Signal/Mode	None
TR	2530.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI	1100 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	60 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tfl_me
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Reordering	Linear
Bandwidth 1	650 Hz/Px
Bandwidth 2	650 Hz/Px
Bandwidth 3	650 Hz/Px

Sequence - Part 1

Bandwidth 4	650 Hz/Px
Echo Spacing	9.84 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Readout trajectory	Bipolar
Gradient spoiling	Siemens
Gradient moment factor	1.00
Averaging	RMS

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\Research\Investigators\XChen\31P CSI 20230503 NOE\fastestmap

TA: 8 sec Coil Selection: Manual Vol: 50×28×50 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

Routine

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm
TR	2800.0 ms
TE	50.00 ms
Averages	1
Coil Elements	HC3-6

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	90.00 deg
F >> H	28 mm
R >> L	50 mm
A >> P	50 mm
Reset	Off

Contrast - Common

TR	2800.0 ms
TE	50.00 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm

Geometry - AutoAlign

AutoAlign	---
-----------	-----

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2800.0 ms

Sequence - Common

Sequence Name	fastmp
---------------	--------

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Linear 3-proj
Vol fit factor	100.00 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	4

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\fastestmap

TA: 17 sec Coil Selection: Manual Vol: 50x28x50 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

Routine

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm
TR	2800.0 ms
TE	50.00 ms
Averages	1
Coil Elements	HC3-6

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	90.00 deg
F >> H	28 mm
R >> L	50 mm
A >> P	50 mm
Reset	Off

Contrast - Common

TR	2800.0 ms
TE	50.00 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm

Geometry - AutoAlign

AutoAlign	---
-----------	-----

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2800.0 ms

Sequence - Common

Sequence Name	fastmp
---------------	--------

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
Vol fit factor	100.00 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	4

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\fastestmap

TA: 17 sec Coil Selection: Manual Vol: 50×28×50 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

Routine

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm
TR	2800.0 ms
TE	50.00 ms
Averages	1
Coil Elements	HC3-6

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	90.00 deg
F >> H	28 mm
R >> L	50 mm
A >> P	50 mm
Reset	Off

Contrast - Common

TR	2800.0 ms
TE	50.00 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm

Geometry - AutoAlign

AutoAlign	---
-----------	-----

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2800.0 ms

Sequence - Common

Sequence Name	fastmp
---------------	--------

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Full 6-proj
Vol fit factor	100.00 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	4

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\fastestmap

TA: 17 sec Coil Selection: Manual Vol: 50x28x50 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

Routine

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm
TR	2800.0 ms
TE	50.00 ms
Averages	1
Coil Elements	HC3-6

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	90.00 deg
F >> H	28 mm
R >> L	50 mm
A >> P	50 mm
Reset	Off

Contrast - Common

TR	2800.0 ms
TE	50.00 ms
Tau	5.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Averages	1
Water Suppression	None

Resolution - Common

Vector Size	256
Normalize	Off

Geometry - Common

Position	L7.4 A35.1 H17.8 mm
Orientation	C > T-17.0
Rotation	0.00 deg
Vol R >> L	50 mm
Vol F >> H	28 mm
Vol A >> P	50 mm

Geometry - AutoAlign

AutoAlign	---
-----------	-----

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2800.0 ms

Sequence - Common

Sequence Name	fastmp
---------------	--------

Sequence - Common

Delta Frequency	0.00 ppm
Measurements	1
Phase Cycling	None
Save Uncombined	Off
Bandwidth	100000 Hz
Acquisition Duration	2 ms

Sequence - Special

Type of fit	Linear 6-proj
Vol fit factor	100.00 %
Force spherical fit Vol	Off Force spherical fit Vol
Save plots to database	Off
Refocus pulses	Normal
Excite pulse duration	6400 us
Refocus pulse duration	6400 us
Bar FoV	384 mm
Bar thickness	10.00 mm
Inversion pulse	Off
Multi-echo acquisition	On
Number of echoes	4

\\Research\Investigators\XChen\31P CSI 20230503 NOE\SPECIAL_ACC

TA: 3:24 min Coil Selection: Auto Vol: 32x20x34 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	C > T-24.6 > S4.6
Rotation	1.26 deg
Vol R >> L	32 mm
Vol F >> H	20 mm
Vol A >> P	34 mm
TR	3000.0 ms
TE	30.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	30.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	64
Water Suppression	Water Saturation
Water Suppr. BW	35 Hz
Spectral Suppr.	None

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	C > T-24.6 > S4.6
Rotation	1.26 deg
Vol R >> L	32 mm
Vol F >> H	20 mm
Vol A >> P	34 mm

Geometry - AutoAlign

AutoAlign	---
-----------	-----

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**System - Miscellaneous**

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	Isocenter
Orientation	C > T-24.6 > S4.6
Rotation	91.26 deg
F >> H	20 mm
R >> L	32 mm
A >> P	34 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Physio - PACE

Resp. Control	Off
---------------	-----

Sequence - Common

Sequence Name	svs_se
Preparation Scans	4
Delta Frequency	0.00 ppm
Ref. Scan Mode	Off
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	1000 Hz
Acquisition Duration	2048 ms
Remove Oversampling	On
Freq. Corr. Accumulation	Off
Incr. Gradient Spoiling	Off

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\head_csi_fid

TA: 36:24 min Coil Selection: Auto Voxel Size: 50.0×50.0×36.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	36 mm
Vol A >> P	400 mm
Vol R >> L	400 mm
TR	3000.0 ms
TE	2.30 ms
Averages	60

Contrast - Common

TR	3000.0 ms
TE	2.30 ms
Flip Angle	90 deg
Preparation Scans	8
Averages	60
Water Suppression	None

Resolution - Common

FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	36 mm
Scan Res. A >> P	8
Scan Res. R >> L	8
Interpol. Res. A >> P	8
Interpol. Res. R >> L	8
Hamming	Off
Dimension	2D
Phase Encoding	Weighted
Vector Size	512
Normalize	Off

Geometry - Common

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	36 mm
Vol A >> P	400 mm
Vol R >> L	400 mm

Geometry - AutoAlign

Slice Group	1
Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
A >> P	400 mm
R >> L	400 mm
F >> H	36 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 31P	49.891823 MHz
Frequency 1H	123.249275 MHz
? Ref. Amplitude 31P	0.000 V
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	csi_fid
Preparation Scans	8
Delta Frequency	0.00 ppm
Measurements	1
Dimension	2D
Save Uncombined	Off
Acquisition Delay	2.30 ms
Bandwidth	4000 Hz
Acquisition Duration	128 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	31P
TX/RX Delta Frequency	0 Hz
NOE Type	Rectangular
NOE Count	1
NOE Duration	5.00 ms
NOE Flip Angle	90.00 deg
TX Nucleus	1H
TX Delta Frequency	0 Hz
Decoupling Type	Waltz4
Decoupling Duration	1.00 ms
DC Total Duration	50.00 %
DC Pause Fract.	100.00 %
Decoupling Flip Angle	180.00 deg

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\head_csi_fid

TA: 14:36 min Coil Selection: Auto Voxel Size: 50.0×50.0×40.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	40 mm
Vol A >> P	400 mm
Vol R >> L	400 mm
TR	3000.0 ms
TE	2.30 ms
Averages	22

Contrast - Common

TR	3000.0 ms
TE	2.30 ms
Flip Angle	90 deg
Preparation Scans	8
Averages	22
Water Suppression	None

Resolution - Common

FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	40 mm
Scan Res. A >> P	8
Scan Res. R >> L	8
Interpol. Res. A >> P	8
Interpol. Res. R >> L	8
Hamming	On
Dimension	2D
Phase Encoding	Weighted
Vector Size	512
Normalize	Off

Geometry - Common

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
FOV A >> P	400 mm
FOV R >> L	400 mm
Thickness F >> H	40 mm
Vol A >> P	400 mm
Vol R >> L	400 mm

Geometry - AutoAlign

Slice Group	1
Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	L30.0 A10.7 H24.3 mm
Orientation	T > C-17.0
Rotation	0.00 deg
A >> P	400 mm
R >> L	400 mm
F >> H	40 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 31P	49.891823 MHz
Frequency 1H	123.249275 MHz
? Ref. Amplitude 31P	0.000 V
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	csi_fid
Preparation Scans	8
Delta Frequency	0.00 ppm
Measurements	1
Dimension	2D
Save Uncombined	Off
Acquisition Delay	2.30 ms
Bandwidth	4000 Hz
Acquisition Duration	128 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	31P
TX/RX Delta Frequency	0 Hz
NOE Type	Rectangular
NOE Count	10
NOE Duration	5.00 ms
NOE Pause	10.00 ms
NOE Flip Angle	90.00 deg
TX Nucleus	1H
TX Delta Frequency	0 Hz
Decoupling Type	Waltz4
Decoupling Duration	1.00 ms
DC Total Duration	50.00 %
DC Pause Fract.	100.00 %
Decoupling Flip Angle	180.00 deg

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\Ax TSE PD T2

TA: 3:32 min Coil Selection: Auto Voxel Size: 0.4×0.4×4.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Multiple Series	Each Measurement
-----------------	------------------

Resolution - Common

FOV Read	210 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
Base Resolution	256
Phase Resolution	80 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off

Routine

Slice Group	1
Slices	30
Distance Factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	3000.0 ms
TE 1	9.00 ms
TE 2	81.00 ms
Averages	1
Concatenations	2
AutoAlign	Head > Brain

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	3000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	2

Contrast - Common

TR	3000.0 ms
TE 1	9.00 ms
TE 2	81.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	120 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	2
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F37.0
L	0.0 mm
P	0.0 mm
H	37.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	37 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	171 mm
A >> P	210 mm
F >> H	149 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms
Concatenations	2

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	210 mm
FOV Phase	81.3 %
Phase Resolution	80 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	2

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Motion Correction	None
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tse
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	250 Hz/Px
Echo Spacing	8.96 ms
Free Echo Spacing	Off
Define	Turbo Factor
Turbo Factor	5
Echo Trains per Slice	34

Sequence - Part 2

Introduction	On
Phase Correction	Automatic
Compensate T2 Decay	Off
Fast Mode	Off
WARP	Off
Red. EC Sensitivity	Off
Acoustic noise reduction	Off

Sequence - Part 2

Reduce Motion Sens.	On
Motion Correction	None

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	130 deg
Allowed Delay	60 s

\\Research\Investigators\XChen\31P CSI 20230503 NOE\t2_tse_dark-fluid_tra

TA: 4:50 min Coil Selection: Auto Voxel Size: 0.8×0.8×4.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	30
Distance Factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	9000.0 ms
TE	114.00 ms
Averages	1
Concatenations	2
AutoAlign	Head > Brain

Contrast - Common

TR	9000.0 ms
TE	114.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	Slice-sel. IR
TI	2500 ms
Freeze Suppr. Tissue	Off
Flip Angle Mode	Constant
Flip Angle	130 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	210 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
Base Resolution	256
Phase Resolution	90 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	25 %
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	9000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	2

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F37.0
L	0.0 mm
P	0.0 mm
H	37.0 mm
Initial Orientation	Transversal

Geometry - AutoAlign

Initial Rotation	90.00 deg
------------------	-----------

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	37 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P0.0 H0.0 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	171 mm
A >> P	210 mm
F >> H	149 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
-----------------	------

Physio - Signal

TR	9000.0 ms
Concatenations	2

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	Slice-sel. IR
TI	2500 ms
Dark Blood	Off
FOV Read	210 mm
FOV Phase	81.3 %
Phase Resolution	90 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	2

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Motion Correction	None
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Sequence - Part 1

Sequence Name	tir
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	179 Hz/Px
Echo Spacing	10.3 ms
Free Echo Spacing	Off
Define	Turbo Factor
Turbo Factor	13
Echo Trains per Slice	15

Sequence - Part 2

Introduction	On
Phase Correction	Automatic

Sequence - Part 2

Compensate T2 Decay	Off
Fast Mode	Off
WARP	Off
Red. EC Sensitivity	Off
Acoustic noise reduction	Off
Reduce Motion Sens.	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	60 s

\\Research\\Investigators\\XChen\\31P CSI 20230503 NOE\\resolve_4scan_trace_tra_p2_192

TA: 2:34 min Coil Selection: Auto Voxel Size: 1.1×1.1×4.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	30
Distance Factor	25 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	217 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	5870.0 ms
TE 1	82 ms
TE 2	151 ms
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	5870.0 ms
TE 1	82 ms
TE 2	151 ms
MTC	Off
Magn. Preparation	None
Flip Angle	180 deg
Fat-Water Contrast	Fat Saturation
Fat Saturation	Strong
Contrasts	2
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Resolution - Common

FOV Read	217 mm
----------	--------

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	192
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Accel. Mode	None
Phase Partial Fourier	Off
Readout Partial Fourier	5/8
Readout Segments	7

Resolution - Filter

Raw Filter	On
Distortion Correction	Off
Normalize	Prescan
Noise Masking	Off

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	25 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	217 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	5870.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F37.0
L	0.0 mm
P	0.0 mm
F	37.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	217 mm
R >> L	217 mm
F >> H	149 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	123.249275 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5870.0 ms
Concatenations	1

Diff

Diffusion Mode	4-Scan Trace
Diff. Directions	4
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²

Diff

b-value 2	1000 s/mm ²
Averages 1	1
Averages 2	1
Invert Gray Scale	Off
Diff. Weighted Images	Off
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	On
Exponential ADC Maps	Off
b-value >=	0 s/mm ²
ADC Noise Threshold	100
Noise Masking	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	resolve
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	723 Hz/Px
Echo Spacing	0.32 ms
Optimization	Min. TE
EPI Factor	192

Sequence - Part 2

Introduction	On
Reacquisition Mode	On

Sequence - Assistant

SAR Assistant	Off
Optimization	Min. TE